



New York University
Computer Science Department
Introduction to Software Development

Dr. Nader Nassar

Homework 3: *Practicing control statements and Strings.*

Deadline: *See NYUclasses for the deadline, 5% off per day after the deadline (3 days maximum).*

Please read the guidelines carefully to avoid receiving a zero on the HW:

- Please use the following naming convention “ExerciseX.java”, where X is the number of the exercise.
- Apply this convention to all future homework assignments during the semester.
- Compile and run the program for each exercises.
- Make an archive (zip file or compressed file) with all the java files (the .java files NOT the .class files) and post it on NYU Classes (StudentName_Homework_X.zip)
- It is your responsibility to make sure that the Zip files have your actual files. You may send the file to yourself by email to double check before you upload on NYU classes.
- If the graders cannot open the file, you will receive a grade of zero.
- If you send the .class files instead of the .java files (source files) you will receive a zero.
- Cheating will be severely addressed with an immediate zero on the homework and a report to the academic advisor and the administration.
- You will automatically lose 50% of the points for an exercise if the program does not compile and run correctly

Exercise 1 (10pts): Write a computer program that asks the user for a password and checks if the password is correct. If the password is valid, you display a message to the user saying “Valid Password”. Otherwise, you display the message “Invalid Password”.

Assume that the valid password is “NYU42day”

Exercise 2 (10pts): Write a Java program that computes the maximum of two given numbers using Java.Lang.Math. You need to read the numbers from the console and display the maximum to the console.

Exercise 3 (10pts): Write a Java program to ask the user to enter letter “a” to generate a *random char* (a-z), and ask the user to enter letter “A” to generate a random char between (A-Z). If the user enters anything else other than “a” or “A” the program will print message, invalid entry, please try again

Example:

Enter letter a to generate a *random char* (a-z), or enter letter “A” to generate a random char between (A-Z).

X

invalid entry, please try again

A

Your random chat is L.

Exercise 4 (15pts): Write a Java program that calculates the letter grade given the numeric percent grade as an

input by user. Your output will look like the following

You entered [numeric grade], Your letter grade is [letter grade]

Use the grading scale we introduced in the first lecture for this class.

Example

Please enter numeric percentage:

85

You entered 85%, your letter grade is B

Exercise 5 (15pts): Write a java program that ask the user to enter the month name in any case (all caps, all small caps, mix caps) and the program will output the season.

Hint: use if/else if

input:

Enter month name: January (user can enter case insensitive month name like jAnuaRy or JanUary, ...etc))

Output:

Season is: Winter

Exercise 6 (20 pts): Write a Java program that examine the user's password strength as follows:

The program will ask the user to enter password.

Case #	Password rule	program output	Example
Case 1	Password length is less that 8 characters (any character combination).	Weak Password	&g87A
Case 2	Password is 8-12 characters all numbers or all alphabets	Medium Password	24730657739 or uuhUWhCQEL
Case 3	Password length is 8-12: all are small caps alphabet characters with more than one digit, or all upper caps alphabet characters with more	OK Password	b0ma9wdelmud or 8MIWRT3OJ7S1

	than one digit.		
Case 4	Password is 8-12: mix caps alphabet characters and at least one digit	Strong Password	71VS3aDgFkTe
Case 5	Password is more than 12 characters (any type of characters)	Awesome password	jR^x~u#ig@h0*{
Case 6	If anything else outside the above conditions	Invalid Password	

Exercise 7 (20 pts): Design a simple ATM service kiosk. This kiosk supports the following options for its menu:

1. View your account balance
2. Deposit Cash
3. Withdraw cash
4. Exit

Enter your selection:

Your code should handle the following use cases:

- If the user enters a selection that is not between 1-4, the ATM displays the message: “Incorrect selection, please try again”, then it displays the four selections again.
- Initially, the account balance is \$0.00. When you deposit money, this account balance is increased by your deposit amount.
- When you withdraw cash, this account balance is decreased by your withdrawal amount then displays the menu again.
- You cannot withdraw more than what exists in the account, otherwise the ATM machine will display an error message “Sorry, you don’t have sufficient balance!” then displays the menu again.
- You cannot withdraw negative cash amount, otherwise the ATM machine will display an error message “Incorrect Amount!” then displays the menu again.
- Similarly, you cannot deposit a negative cash amount, otherwise the ATM displays the message “Incorrect Amount!” then displays the menu again.
- If the user enters 4, the ATM displays “Goodbye” and the program exists

1. View your Balance
2. Deposit Cash
3. Withdraw Cash
4. Exit

Enter your selection: 1

Your current balance is 0.00

1. View your Balance
2. Deposit Cash
3. Withdraw Cash
4. Exit

Enter your selection: 2

Enter the amount you want to deposit: 100

1. View your Balance
2. Deposit Cash
3. Withdraw Cash
4. Exit

Enter your selection: 3

Enter the amount your want to withdraw: 97

1. View your Balance
2. Deposit Cash
3. Withdraw Cash
4. Exit

Enter your selection: 1

Your current balance is 3.00

1. View your Balance
2. Deposit Cash
3. Withdraw Cash
4. Exit

Enter your selection: 3

Enter the amount your want to withdraw: 5

Sorry you don't have enough balance!

1. View your Balance
2. Deposit Cash
3. Withdraw Cash
4. Exit

Enter your selection: 4

Goodbye